

D 3 A

$$y = \lim_{n \rightarrow +\infty} \frac{5^n - n^2 4^n}{n 2^n - 3^n} \approx \frac{-n^2 4^n}{n 2^n} = \frac{-n 4^n}{2^n} = -\infty$$

D 6

$$y = x^2 e^{2x-1}$$

$$y' = (2x)(e^{2x-1}) + (x^2)([e^{2x-1}][2]) = 2x e^{2x-1} + 2x^2 e^{2x-1} = 2e^{2x-1}(x + x^2)$$

$$2e^{2x-1}(x + x^2) \leq 0 \quad \left\{ \begin{array}{l} e^{2x-1} \leq 0 \\ x(1+x) \leq 0 \end{array} \right. \quad \left\{ \begin{array}{l} x \leq 0 \\ x+1 \leq 0 \end{array} \right. \quad \left\{ \begin{array}{l} x \leq 0 \\ x \leq -1 \end{array} \right. \quad \begin{array}{c} \frac{-1}{x} \quad \frac{0}{-1} \\ \hline -1 \quad + \end{array} \quad x \in [-1, 0]$$

D 8

$$\int_0^K \frac{x}{x^2+1} = \left[\frac{1}{2} \ln(x^2+1) \right]_0^K = \left(\frac{1}{2} \ln(K^2+1) \right) - \left(\frac{1}{2} \ln(1) \right) = \frac{1}{2} \ln(K^2+1) - 0 = \frac{1}{2} \ln(K^2+1) = \frac{\ln(2)}{2}$$

$$\ln(K^2+1) = \ln(2)$$

$$K^2+1 = 2$$

$$K^2 = 1$$

$$K = \pm 1$$

$$\int \frac{x}{x^2+1} = \int \frac{\cancel{x}}{x^2+1} \cdot \frac{1}{\cancel{2x}} = \frac{1}{2} \int \frac{1}{y} = \frac{1}{2} \ln(y) = \frac{1}{2} \ln(x^2+1)$$

$$y = x^2 + 1$$

$$dy = 2x dx$$

$$\frac{dy}{2x} = dx$$

D 2

$$\ln(x+1) > 2 \ln(x) \quad x \in (0, (1+\sqrt{5})/2)$$

$$\ln(x+1) > \ln(x^2)$$

$$x+1 > x^2$$

$$-x^2 + x + 1 > 0$$

$$x^2 - x - 1 < 0$$

D 5

$$y = \sin(\pi/x)$$

$$y' = \cos(\pi/x) \cdot \frac{-(\pi)}{x^2} = \cos(\pi/x) \cdot \frac{(-\pi)}{x^2} = -\frac{\pi \cos(\pi/x)}{x^2}$$

$$y'(1) = -\frac{\pi \cos(\pi)}{1} = \frac{-\pi \cdot -1}{1} = \pi$$

D 7

$$2(\ln(x+1) + x^2) = y$$

$$y' = \frac{2}{x+1} + 2x$$

$$y'' = \frac{-(2)(1)}{(x+1)^2} + 2 = -\frac{2}{(x+1)^2} + 2$$

$$-\frac{2}{(x+1)^2} + 2 = 0$$

$$-\frac{2}{(x+1)^2} = -2 \quad \frac{2}{(x+1)^2} = 2 \quad \frac{1}{(x+1)^2} = 1 \quad (x+1)^2 = 1$$

$$x^2 + 2x + 1 = 1$$

$$x^2 + 2x = 0$$

$$x(x+2) = 0 \Rightarrow x = 0 \vee x = -2$$

D 4

$$y = \begin{cases} \frac{\ln(1-2x)}{x+x^2} \\ 2ae^x + \cos x \end{cases}$$

$$\frac{-1}{1+2x} \cdot 2 = \frac{-2}{1+2x} \quad \lim_{x \rightarrow 0^+} f(x) = -2$$

$$\lim_{x \rightarrow 0} 2ae^0 + \cos x = 2a + 1$$

$$2a + 1 = -2$$

$$2a = -3$$

$$a = -\frac{3}{2}$$

Eg 1

$$y = x^3 e^{1/x}$$

DOMINIO

$$x \neq 0$$

$$x \in \mathbb{R} \setminus \{0\}$$

SEGNO

$$x^3 e^{1/x} > 0 \quad \left\{ \begin{array}{l} x^3 > 0 \\ e^{1/x} > 0 \end{array} \right\} \quad \left\{ \begin{array}{l} x > 0 \\ \forall x \end{array} \right.$$

$$y > 0 \quad \text{per } x \in (0, +\infty)$$

$$y < 0 \quad \text{per } x \in (-\infty, 0)$$

DERIVATA

$$y = x^3 e^{1/x}$$

$$y' = (3x^2)(e^{1/x}) + (x^3)\left[e^{1/x} \cdot -\frac{1}{x^2}\right] = 3x^2 e^{1/x} + \left(\cancel{x^3} \cdot -\frac{e^{1/x}}{\cancel{x^2}}\right) = 3x^2 e^{1/x} - \frac{e^{1/x}}{x} = e^{1/x}(3x - 1)$$

MONOTONIA

$$e^{1/x}(3x - 1) > 0$$

MAX & MIN

$$e^{1/x}(3x - 1) = 0$$

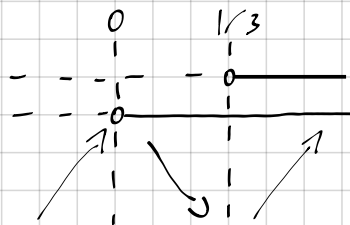
$$3x - 1 = 0$$

$$x = 1/3$$

$f(1/3)$ è minimo locale

$$f(1/3) = e^3/27$$

$$\left\{ \begin{array}{l} x > 0 \\ e^{1/x} > 0 \\ 3x - 1 > 0 \end{array} \right\} \quad \left\{ \begin{array}{l} x > 0 \\ \forall x \\ 3x > 1 \end{array} \right\} \quad \left\{ \begin{array}{l} x > 0 \\ \forall x \\ x > 1/3 \end{array} \right.$$



LIMITI

$$\lim_{x \rightarrow -\infty} x^3 e^{1/x} = -\infty \cdot 1 = -\infty \quad \text{asymptotico obliquo?}$$

$$\lim_{x \rightarrow -\infty} \frac{x^3 e^{1/x}}{x} = x^2 e^{1/x} = -\infty \cdot 1 \quad \text{no as obliquo}$$

Stessa cosa per $+\infty$

$$\lim_{x \rightarrow 0^+} x^3 e^{1/x} = 0^+ \cdot e^{+\infty} = 0^+ \cdot 0^+ = 0^+ \quad \text{no as verticale}$$

Eg 2

1)

dato a_n e b_n 2 successioni a termini positivi se $a_n/b_n = l$ ed $l \in (0, +\infty)$ le 2 serie hanno lo stesso carattere

2)

$$a_n = \log\left(1 + \frac{3}{n^2}\right)$$

$$b_n = 1/n^2$$

a_n è sempre maggiore di 0 perché $1 + \frac{3}{n^2} > 1$ ed $b_n(1) = 0$

b_n è una potenza con esponente pari quindi è sempre positivo

$$\lim_{n \rightarrow +\infty} \frac{\log\left(1 + \frac{3}{n^2}\right)}{1/n^2} = \frac{\log(1 + 3/x^2)}{3/x^2} \cdot \frac{3/x^2}{1/x^2} = 3$$

$y = 1/x^2$
 $y \rightarrow 0$ per $x \rightarrow +\infty$

3)

Per il punto 2 sappiamo che $\sum a_n$ e $\sum b_n$ si comportano allo stesso modo

essendo $\frac{1}{n^2}$ una serie armonica con esponente > 1 $\sum b_n$ converge quindi anche $\sum a_n$ converge

4)

$$\sum \sin(n) \log\left(1 + \frac{3}{n^2}\right)$$

le serie si comporta come $\log(1 + 3/n^2)$ che sappiamo converge dal punto 3

$$0 < \sin(n) \log\left(1 + \frac{3}{n^2}\right) < \log\left(1 + \frac{3}{n^2}\right)$$

per il criterio del confronto $\sum \sin(n) \log\left(1 + \frac{3}{n^2}\right)$ converge

Ex 3

$$y = \sin(x) \sqrt{\cos(x)}$$

$$\int \sin(x) \sqrt{\cos(x)} = \int \frac{\sin \sqrt{\cos x}}{-\sin x} = \int -\sqrt{y} = -\int \sqrt{y} = -\frac{2}{3} y^{3/2} = -\frac{2}{3} (\cos x)^{3/2} = -\frac{2}{3} \cos x \sqrt{\cos x} + C = 2$$

$$\begin{aligned} y &= \cos x \\ dy &= -\sin(x) dx \\ \frac{dy}{-\sin x} &= dx \end{aligned}$$

$$\frac{2}{3} \cos(0) \sqrt{\cos(0)} + C = 2 \quad -\frac{2}{3} 1 \cdot 1 + C = 2$$

$$C = \frac{2}{3} + 2$$

$$C = \frac{6+2}{3} = \frac{8}{3}$$

$$\left[-\frac{2}{3} \cos x \sqrt{\cos x} \right]_{-\pi/2}^{\pi/2} = \left(-\frac{2}{3} \cos \pi/2 \sqrt{\cos \pi/2} \right) - \left(-\frac{2}{3} \cos -\pi/2 \sqrt{\cos -\pi/2} \right) = 0 - 0$$

$$-\frac{2}{3} \cos x \sqrt{\cos x} + \frac{8}{3}$$